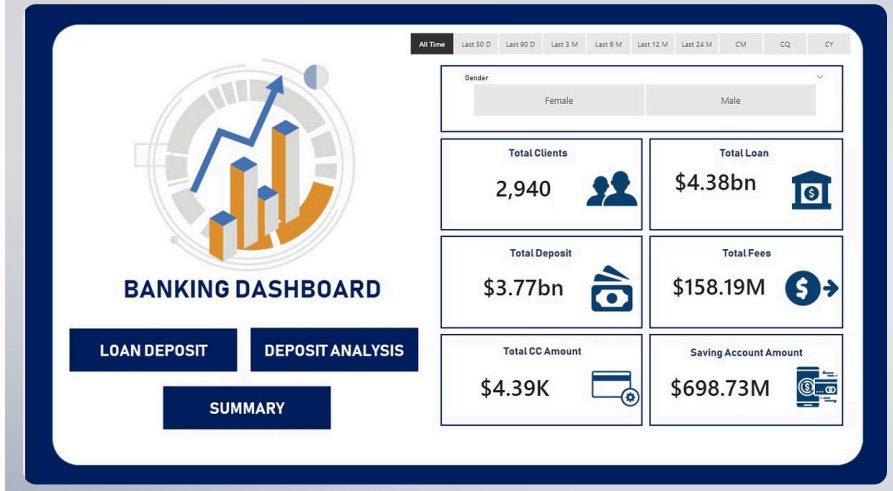
Banking Customer Analytics

Exploratory Data Analysis & Power BI Dashboard

A comprehensive data analytics project examining **2,940 customers** across loans, deposits, demographics, and income segmentation — powered by Python and Power BI.

2,940 Total Customers	$4.38bn Total Loans
$3.77bn Total Deposits	

Presented by: Avi Saini | Data Analytics Project



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Project Overview

Three Pillars of Analysis

This project combines rigorous Python-based exploration with an interactive Power BI dashboard to surface actionable insights across a real-world banking dataset.

EDA with Python

Pandas, Seaborn & Matplotlib used to explore 2,940 customer records — distributions, correlations, and income segmentation across 15+ feature columns.

Power BI Dashboard

4-page interactive dashboard covering Home KPIs, Loan Analysis, Deposit Analysis, and a full Summary view with advisor and gender filters.

Business Insights

Data-driven findings on savings behaviour, loan portfolios, customer demographics, and income-band segmentation to guide strategic decisions.

- Python
- Pandas
- Seaborn
- Matplotlib
- Power BI

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Dataset at a Glance

What the Data Contains

2,940

Total Customers

15+

Feature Columns

6

Product Categories

5

Nationalities

Demographics & Classification

- **Demographics:** Age, Nationality, Occupation, Properties Owned
- **Classification:** Risk Weighting, Loyalty Classification, Income Band, Fee Structure
- **Temporal:** Join Date — parsed to datetime for time-series analysis

Financial Products

- Bank Deposits & Saving Accounts
- Checking Accounts & Foreign Currency
- Business Lending & Bank Loans
- Credit Cards & Engagement Accounts

Income Band engineered via `pd.cut()` on `Banking.csv`

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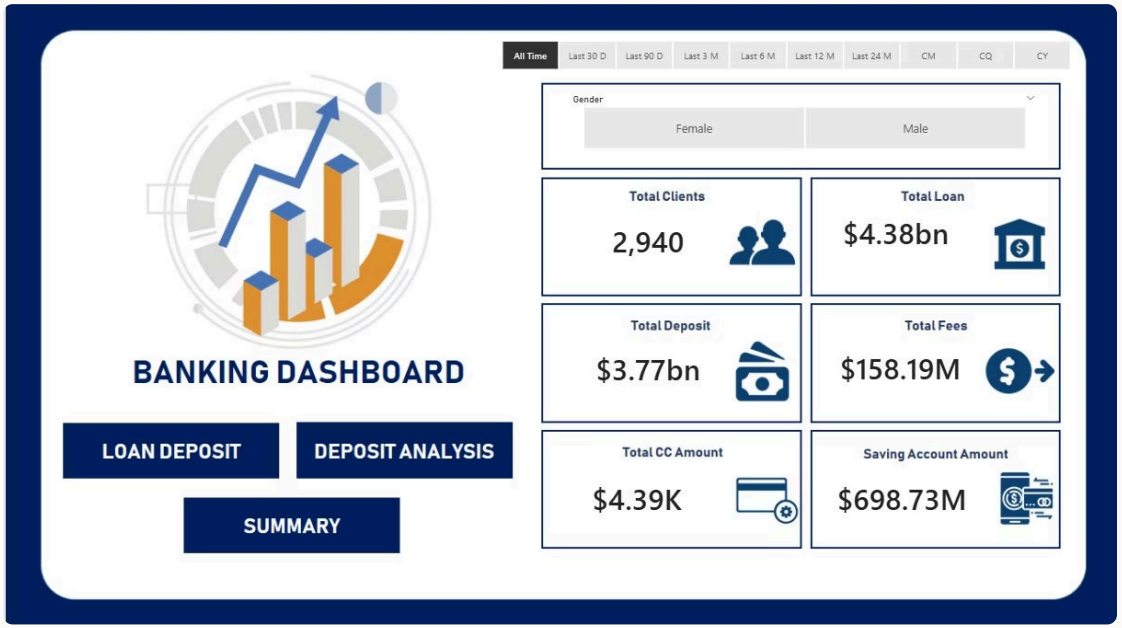
Dashboard Home

Key Portfolio Metrics



2,940
Total Clients

$4.38bn
Total Loans

$3.77bn
Total Deposits

$698.73M
Savings Accounts

$158.19M
Total Fees

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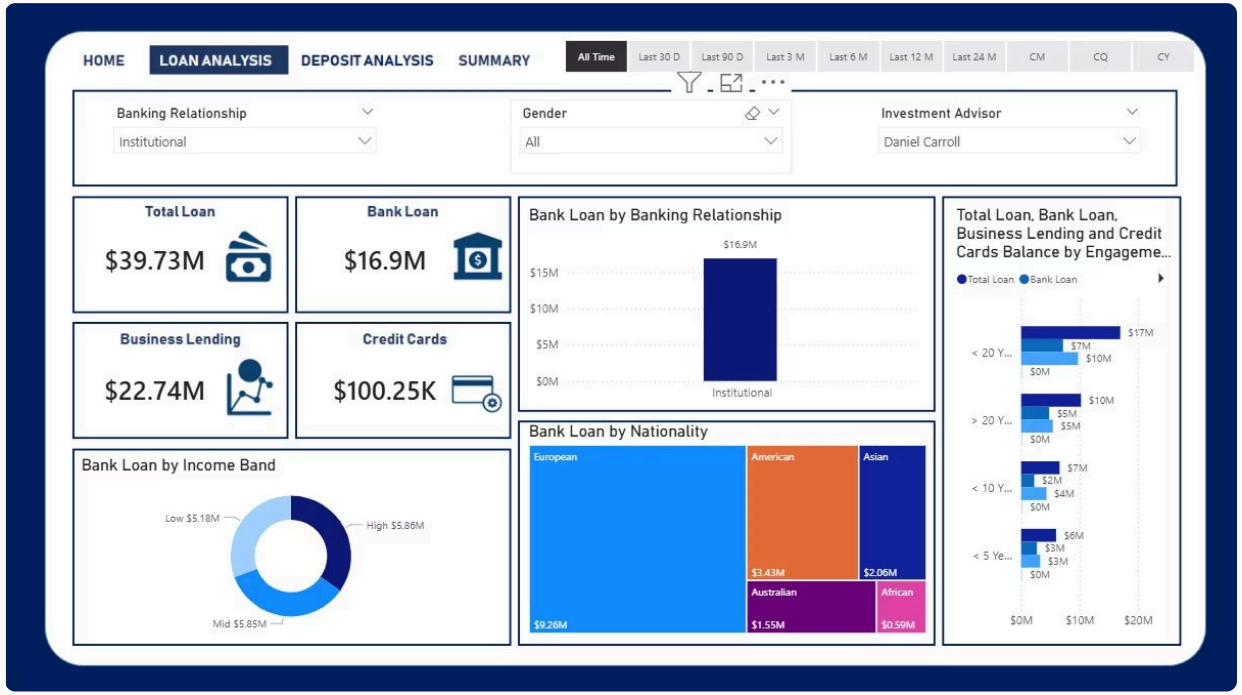
Loan Analysis

Loan Portfolio Breakdown



European customers represent the **largest borrower group** by nationality. Income distribution across loan bands is notably balanced: Mid ($5.85M), High ($5.86M), and Low ($5.18M) — indicating broad credit engagement across all tiers.

$39.73M

Total loan portfolio — Institutional segment

$16.9M

Bank Loan — largest single lending category

$22.74M

Business Lending — dominant across all segments

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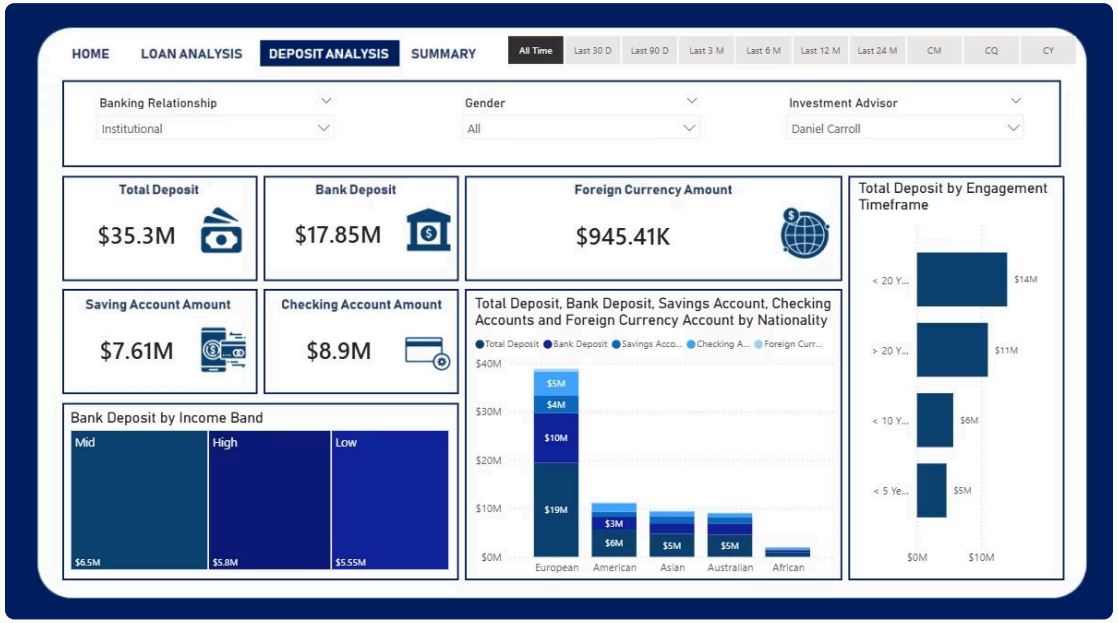
Deposit Analysis

Deposit Portfolio Breakdown



Bank deposits dominate the portfolio at over 50% of total deposits. European and American customers lead by nationality, with deposit engagement rising significantly at longer engagement timeframes.

$35.3M
Total Deposits

$17.85M
Bank Deposits

$7.61M
Saving Accounts

$8.9M
Checking Accounts

$945.41K
Foreign Currency

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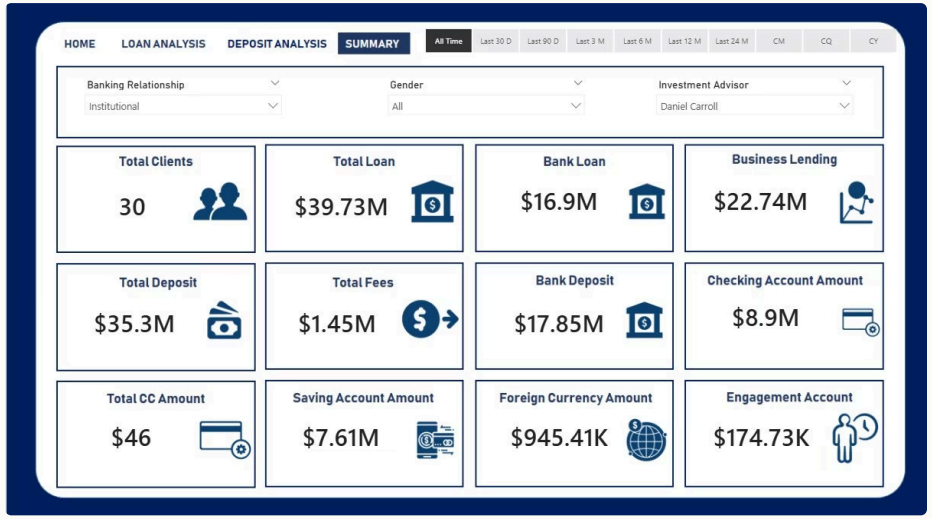
Full Summary View

All KPIs at a Glance



Filtered view: **Institutional** Banking Relationship | Investment Advisor: **Daniel Carroll** | 30 clients in this segment

$39.73M
Total Loans

$35.3M
Total Deposits

$1.45M
Total Fees

$174.73K
Engagement Account

$46
Total CC Amount

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EDA — Key Insights

What the Data Reveals

1

Deposits & Savings Correlation

Bank Deposits and Saving Accounts show **high positive correlation** — customers who actively deposit also tend to grow their savings balances over time.

2

Income & Age Drive Accumulation

Older, higher-income customers hold **significantly larger** superannuation, checking, and savings balances — consistent with financial lifecycle theory.

3

Property Ownership Independent

Properties Owned shows **weak correlation** with banking metrics — driven by external factors like market conditions and inheritance not captured in transaction data.

4

Dual-Debt Customer Segment

Business Lending shows **moderate overlap** with personal Bank Loans — a distinct sub-segment carrying both personal and commercial debt simultaneously.

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Recommendations

Six Strategic Actions

Target High-Value Segment

Focus premium products on older, high-income customers with multi-product holdings and larger savings balances.

Bundle Deposits & Savings

High correlation between deposits and savings supports bundled offerings like auto-sweep accounts to deepen engagement.

Nationality-Based Outreach

European customers dominate loan volumes — multilingual or region-specific products could deepen wallet share.

Dual-Debt Strategy

Tailored relationship manager programs for customers with both business and personal loans — high lifetime value, higher retention risk.

Income Band Monitoring

Mid-income customers show balanced loan exposure. Regular dashboarding by income band enables proactive credit risk management.

Dashboard Automation

Power BI enables real-time filtering by advisor, gender, and relationship — recommend scheduling automated monthly refresh cycles.

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Thank You

Banking Customer Analytics Project

2,940
Clients Analyzed

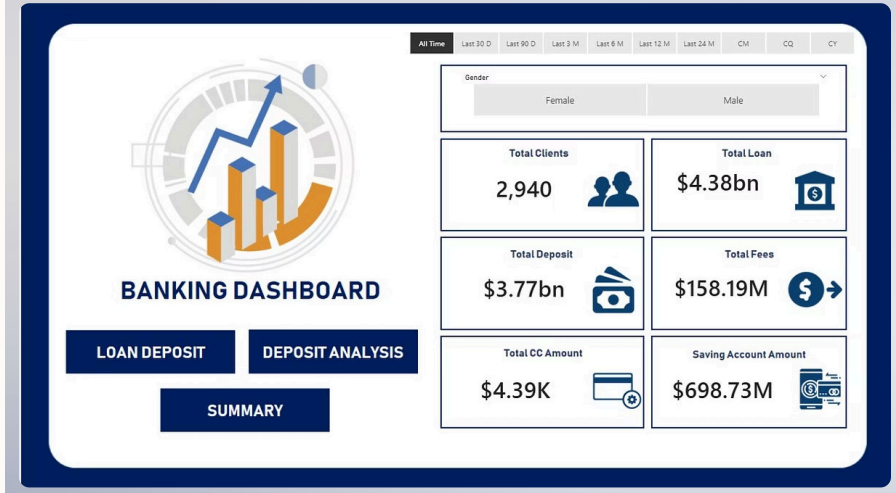
$4.38bn
Loans Tracked

$3.77bn
Deposits Covered

4 Pages
Dashboard Views

👤 **Avi Saini** | 💼 Data Analytics Project

🔗 github.com/avisaini764/banking-customer-analytics



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